ELEC 221 Lecture 17
DT systems based on difference equations; the sampling theorem

Thursday 5 November 2026

Announcements

- No office hours tomorrow
- Reading break: no tutorial Monday, no class Tuesday

Last time

Continuous time:

Discrete time:

Last time

We evaluated the DTFT for a square pulse.

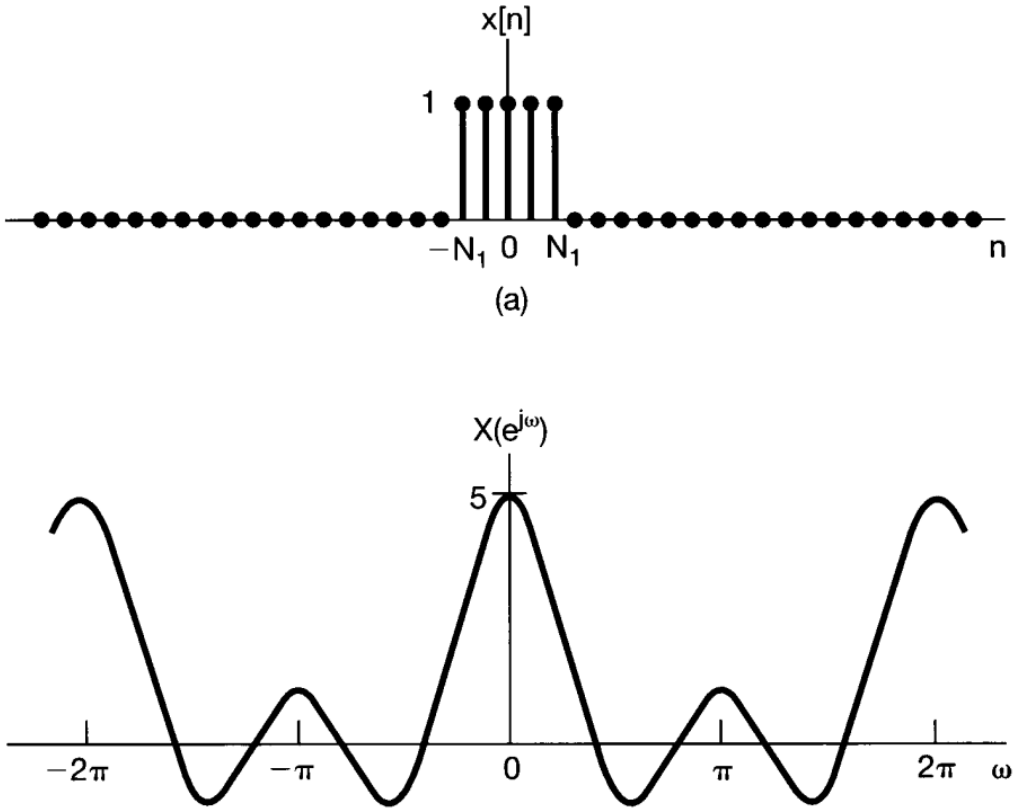
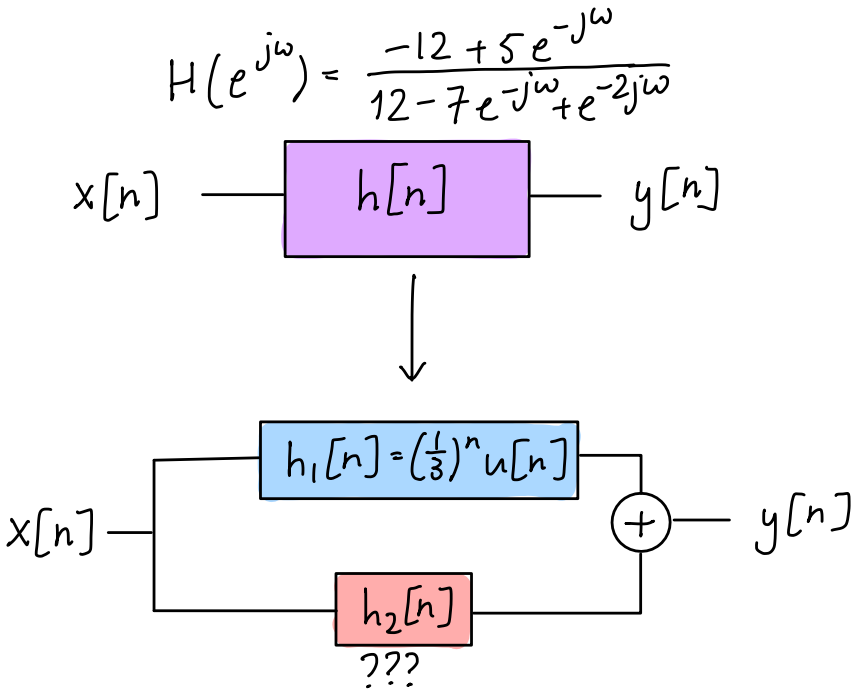


Image credit: Oppenheim chapter 5.1

Last time

We left off with an example of applying the **convolution property** to determine the impulse response of a system.



Can finish solving in this week's tutorial notes

Today

Learning outcomes:

- construct and analyze DT systems based on difference equations
- state the sampling theorem
- define the Nyquist sampling rate and determine if a sampling rate is sufficient to reconstruct a signal from its samples
- describe the phenomenon of aliasing

Properties of the DT Fourier transform

Time shift:

Differencing:

Example: difference equations

What is the frequency response of a system defined by a linear, constant-coefficient difference equation

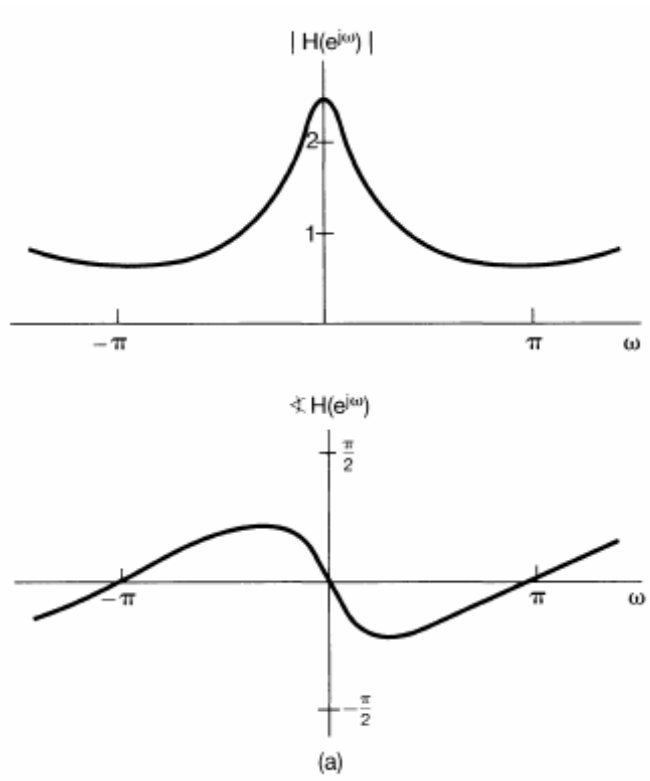
DT systems based on difference equations

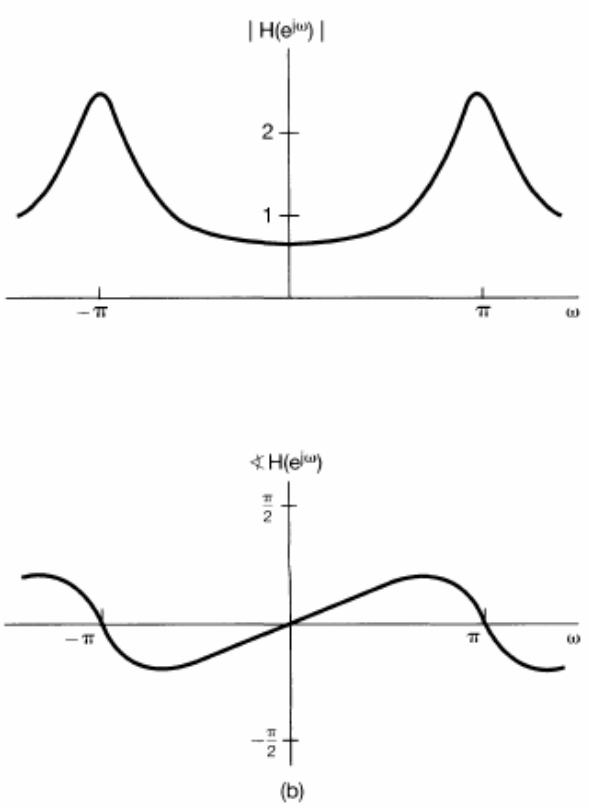
Exercise: compute frequency response of a system characterized by

$$y[n] - ay[n-1] = x[n]$$

Example

"First-order recursive DT filters"





$0 < a < 1 \ (a = 0.6)$

$-1 < a < 0 \ (a = -0.6)$

Image credit: Oppenheim Figure 3.34

DT systems based on difference equations

Exercise: compute frequency response of a system characterized by

$$y[n] = \frac{1}{3}(x[n+1] + x[n] + x[n-1])$$

Can work through the general case in tutorial notes.

DT systems based on difference equations

"Non-recursive DT filters"

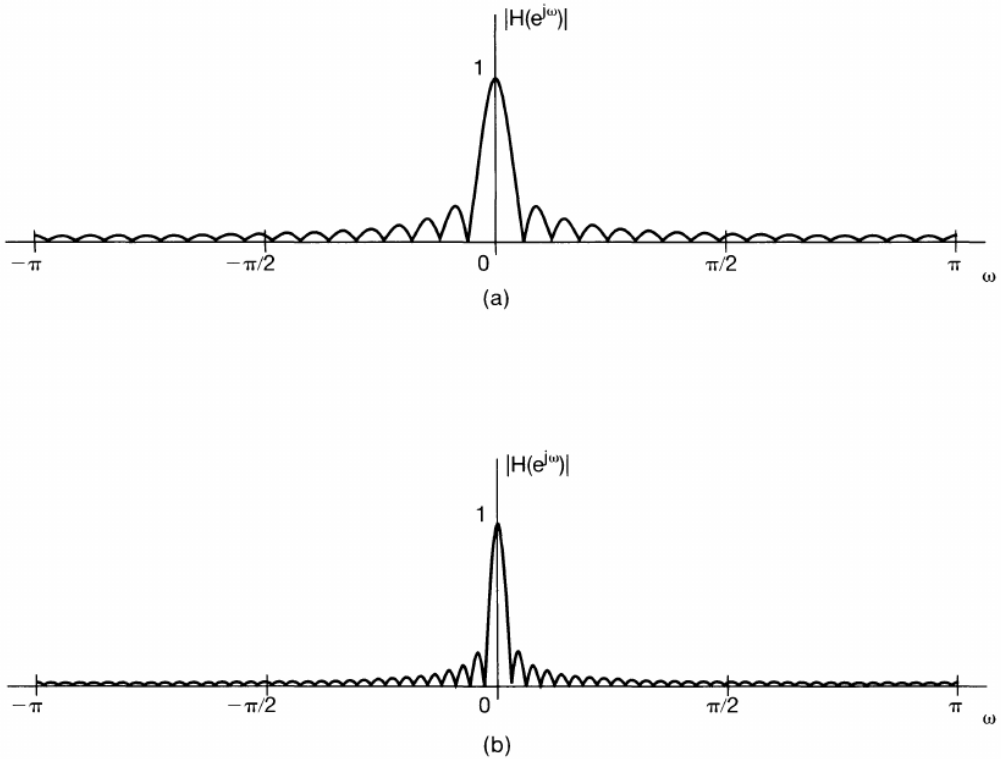


Image credit: Oppenheim Figure 3.36

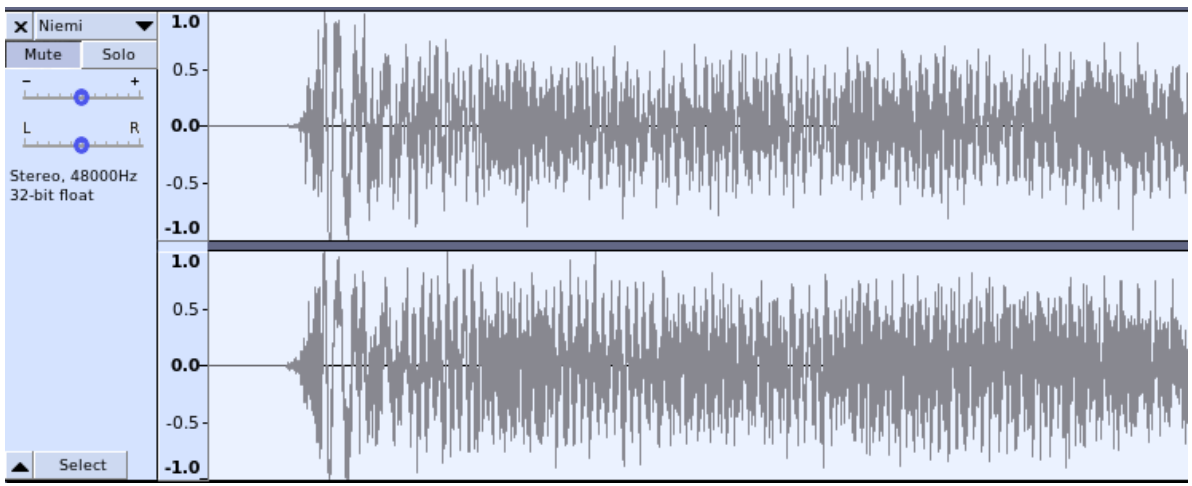
Motivation

Lecture 04 Demos

```
import numpy as np
import matplotlib.pyplot as plt
from IPython.display import Audio
```

Demo 1: fun with square waves

```
tone = 65           # A frequency in Hz
duration = 2         # The length of the audio signal (in seconds)
sample_rate = 48000  # The number of samples per second to take
t_range = np.linspace(0, duration, sample_rate * duration) # Range of time
```



Motivation

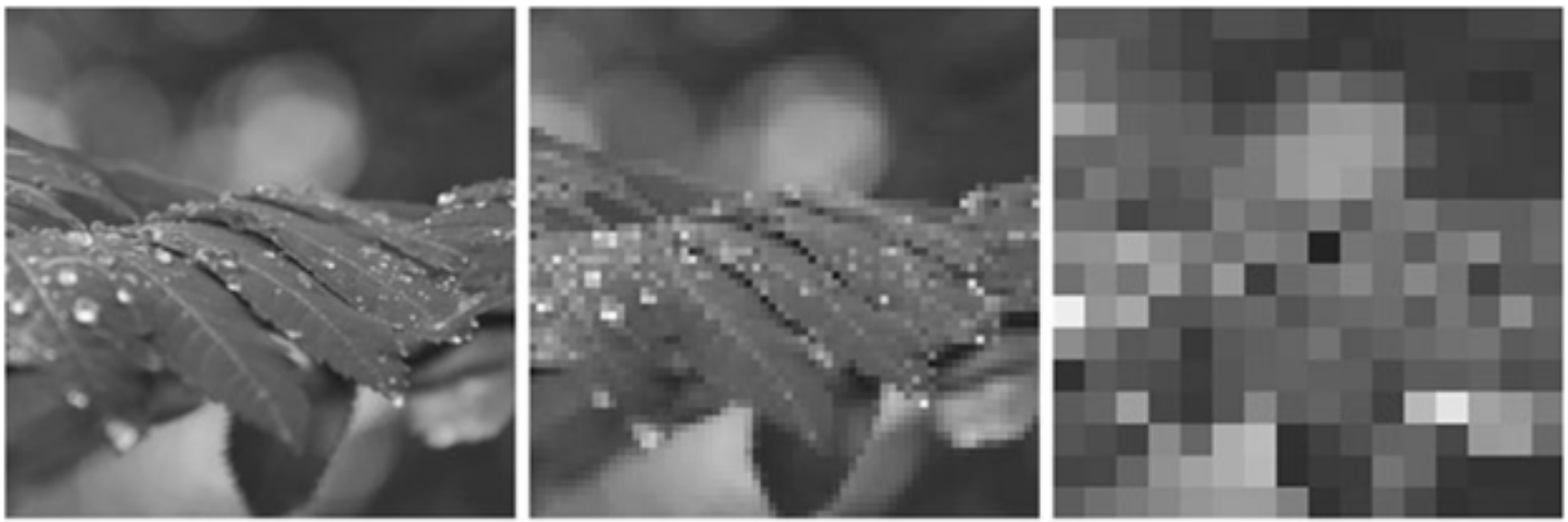


Image credit: `https://what-when-how.com/introduction-to-video-and-image-processing/image-acquisition-introduction-to-video-and-image-processing-part-2/`

Motivation

https://youtu.be/B8EMI3_0T00?t=9



History of frame rate in film:
https://www.youtube.com/watch?v=mjYjFEp9Yx0

Impulse train sampling

Core question: under what conditions can we recover a continuous time signal using only information from its samples?

The unit impulse as a sampler

Multiplying the signal by a shifted impulse picks out the value of the signal at that point:

$$x[n] \cdot \delta[n-k] = x[k] \cdot \delta[n-k]$$

This allows us to write any signal as a **superposition of weighted impulses.**

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot \delta[n-k]$$

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Impulse train sampling

In continuous time:

What if we have more than one?

where

Impulse train sampling

What does the following signal look like?

Impulse train sampling

The combined signal in the time domain is

What happens in the *frequency domain*?

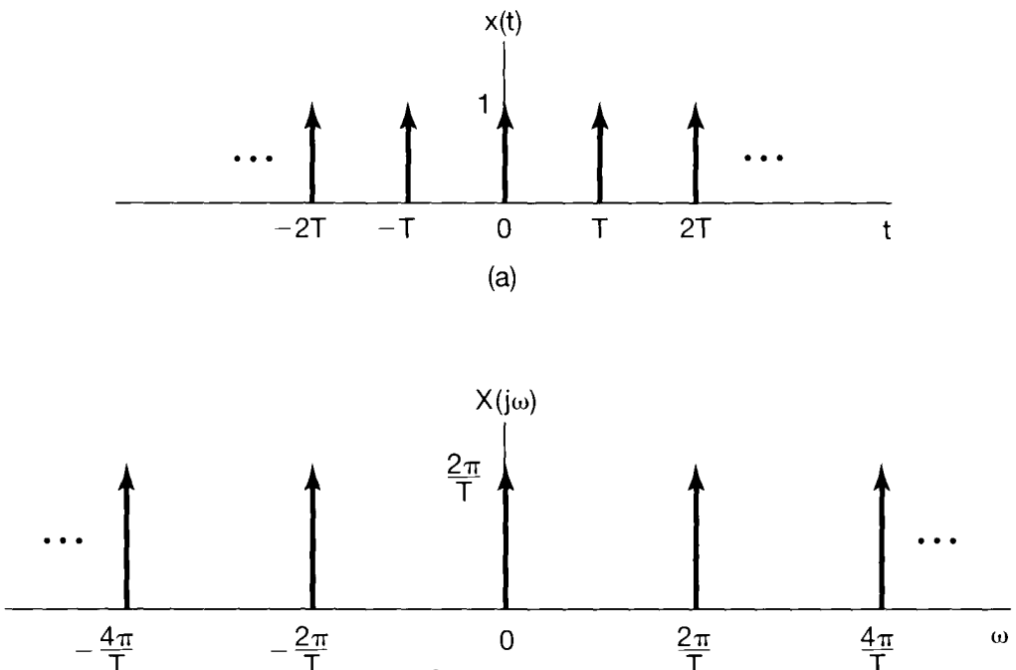
Impulse train sampling

We have a periodic impulse train. Recall what Fourier transforms of periodic signals looked like:

Impulse train sampling

We need to find the Fourier series coefficients of the periodic impulse train.

Impulse train sampling



Impulse train sampling

Impulse train sampling

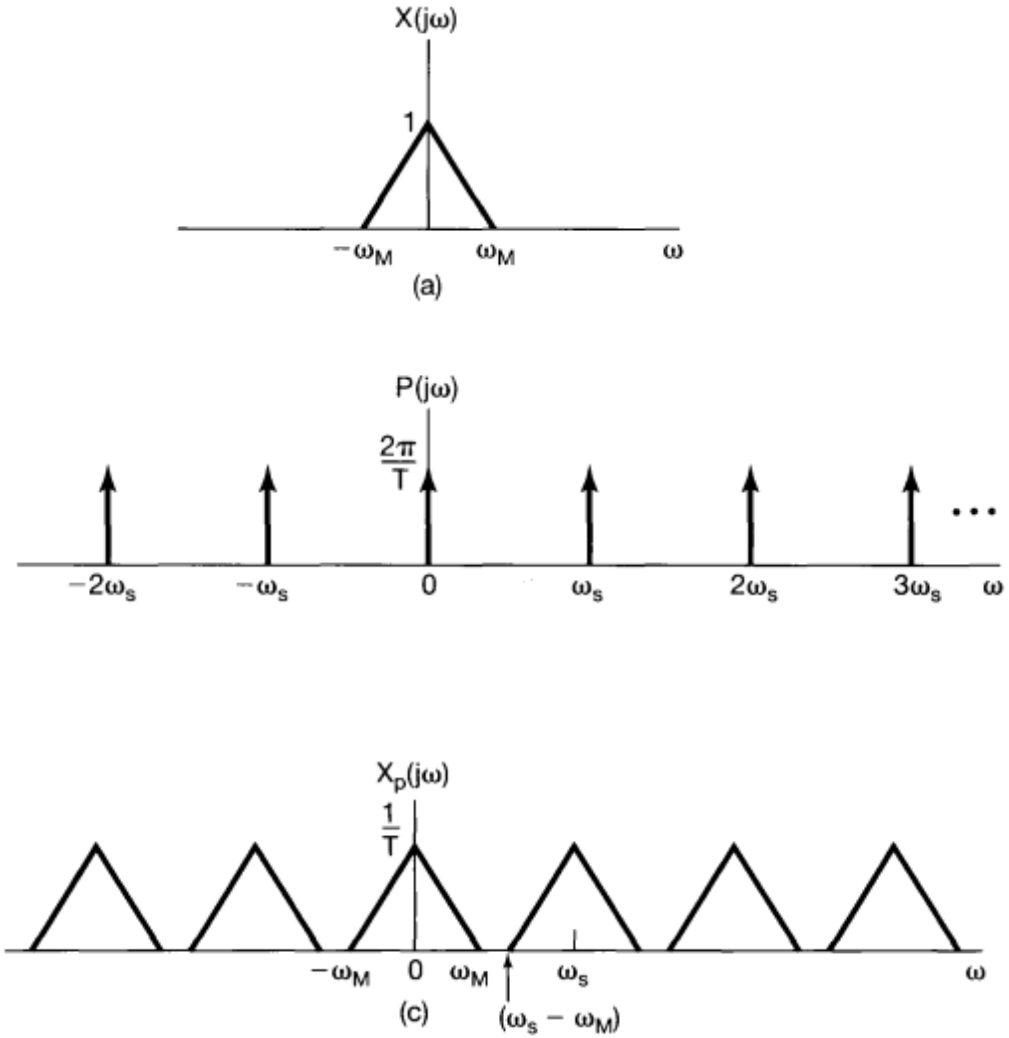
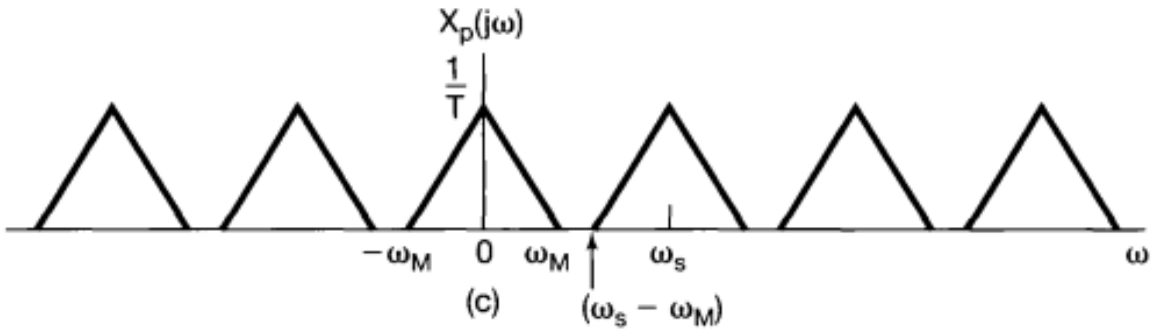


Image credit: Oppenheim 7.1

Impulse train sampling

Suppose we have sampled...



How do we recover our original signal from this spectrum?

Image credit: Oppenheim 7.1

The sampling theorem

"Let $x(t)$ be a **band-limited** signal with $X(j\omega) = 0$ for $|\omega| > \omega_M$. Then $x(t)$ is uniquely determined by its samples $x(nT)$, $n = 0, \pm 1, \pm 2, \ldots$, if

Given these samples, we can reconstruct $x(t)$ by generating a periodic impulse train in which successive impulses have amplitudes that are successive sample values. This impulse train is then processed through an ideal lowpass filter with gain T and cutoff frequency greater than ω_M and less than $\omega_s - \omega_M$. The resulting output signal will exactly equal $x(t)$."

The sampling theorem

Let's show this graphically:

The sampling theorem

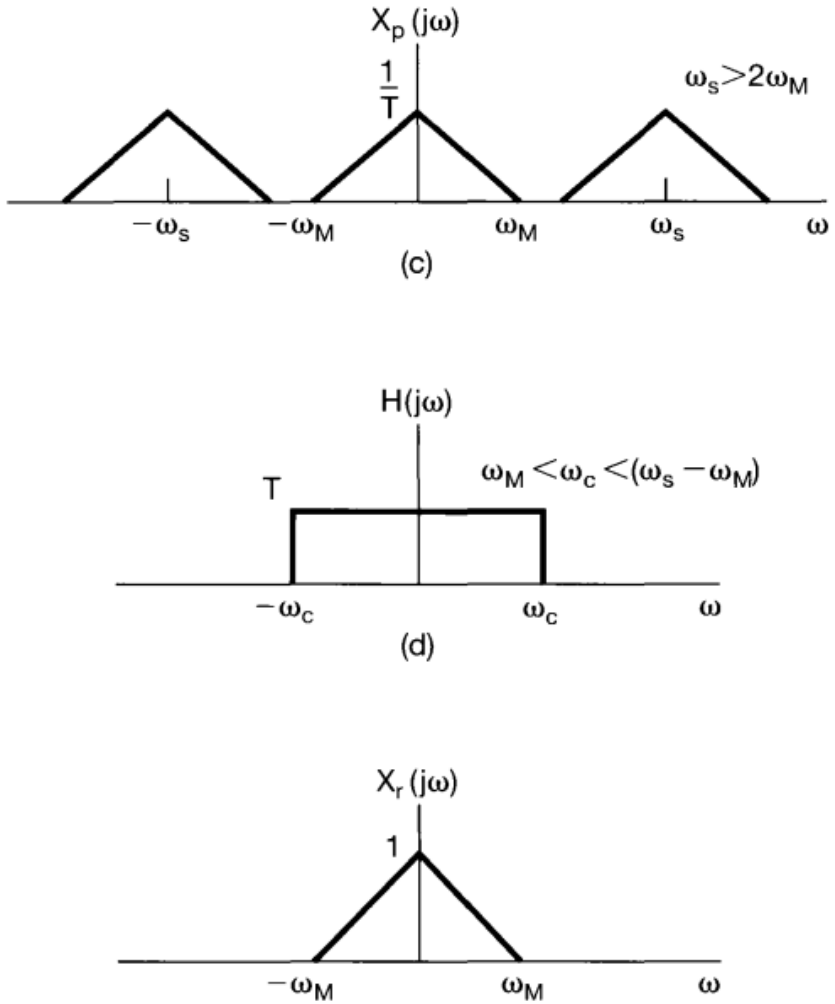


Image credit: Oppenheim 7.1

The Nyquist rate

The sampling frequency is key:

- $\omega_s = 2\omega_M$ is referred to as the **Nyquist rate**
- $\omega_M = \omega_s/2$ is referred to as the **Nyquist frequency**

Exercise: suppose we perform impulse-train sampling with period $T = 10^{-4}$. If a signal $x(t)$ has $X(j\omega) = 0$ for $|\omega| > 15000\pi$, can we reconstruct it exactly from the samples?

Interpolation

In reality we cannot generate a perfect, ideal impulse train. But, we can still interpolate (you will explore this in practice A4)

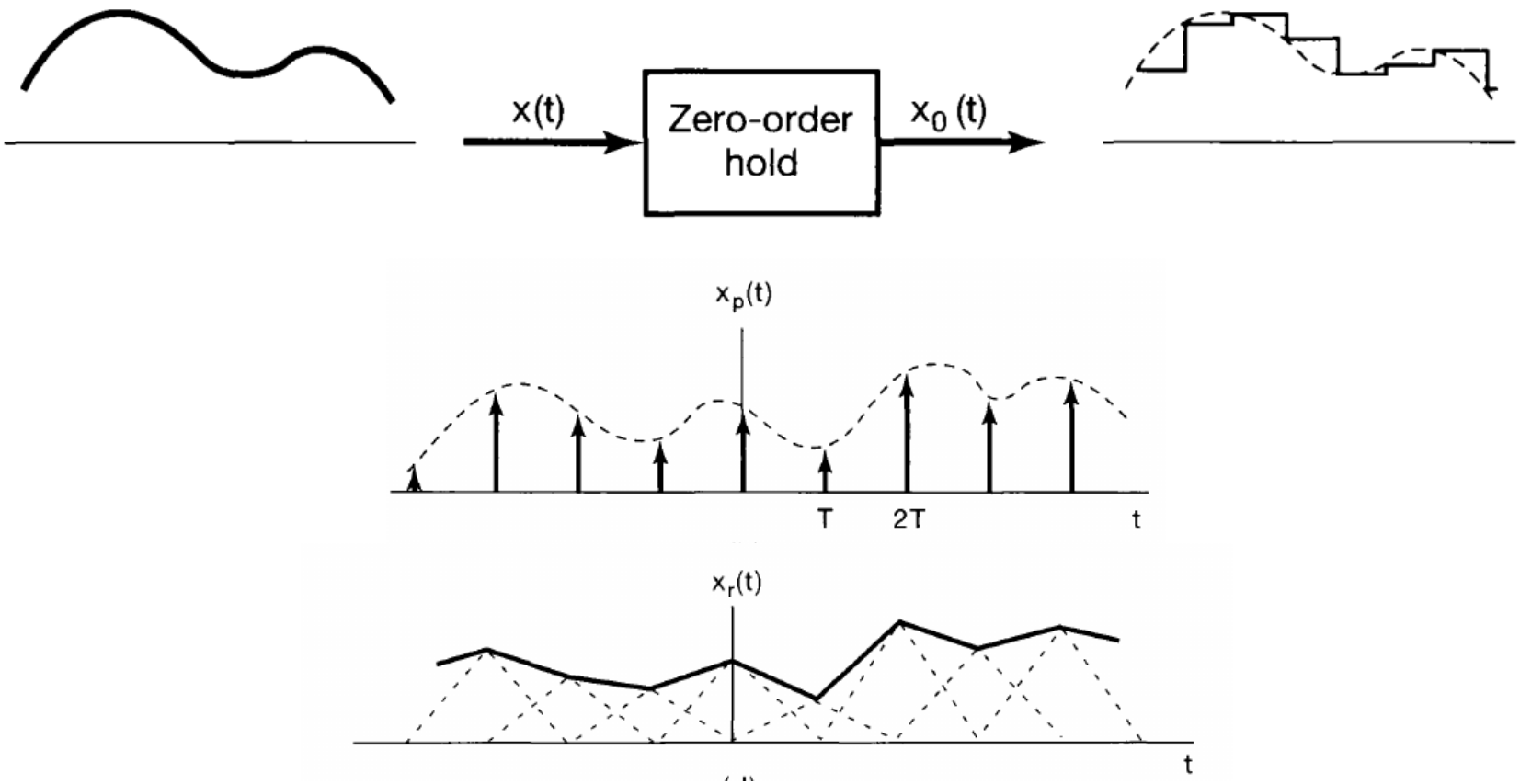


Image credit: Oppenheim 7.1, 7.13

Aliasing

What happens when you don't sample at a high enough rate?

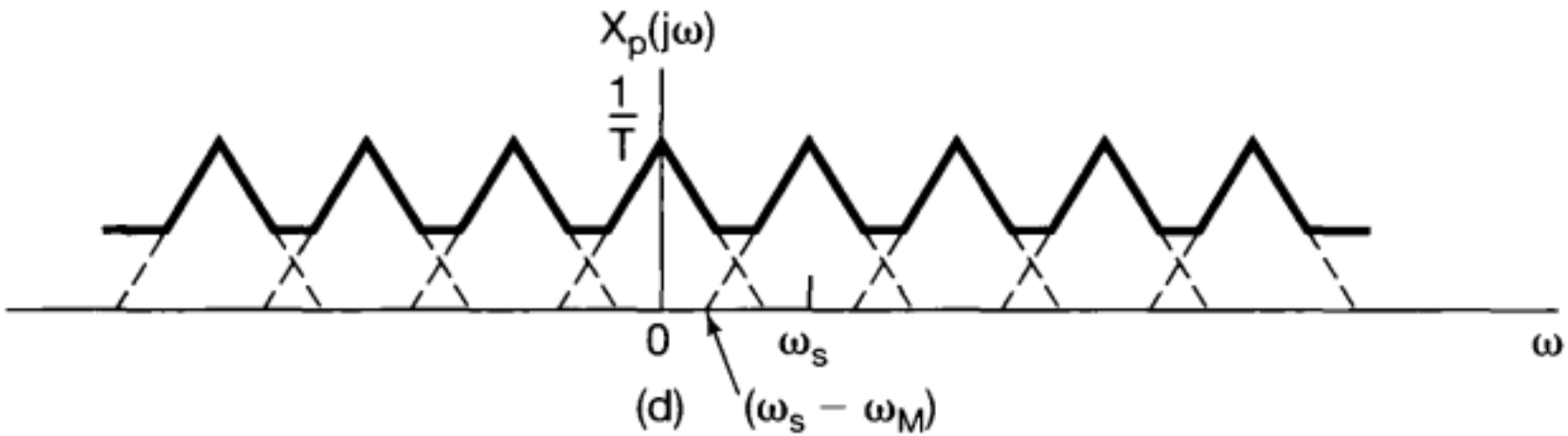


Image credit: Oppenheim 7.1

Aliasing

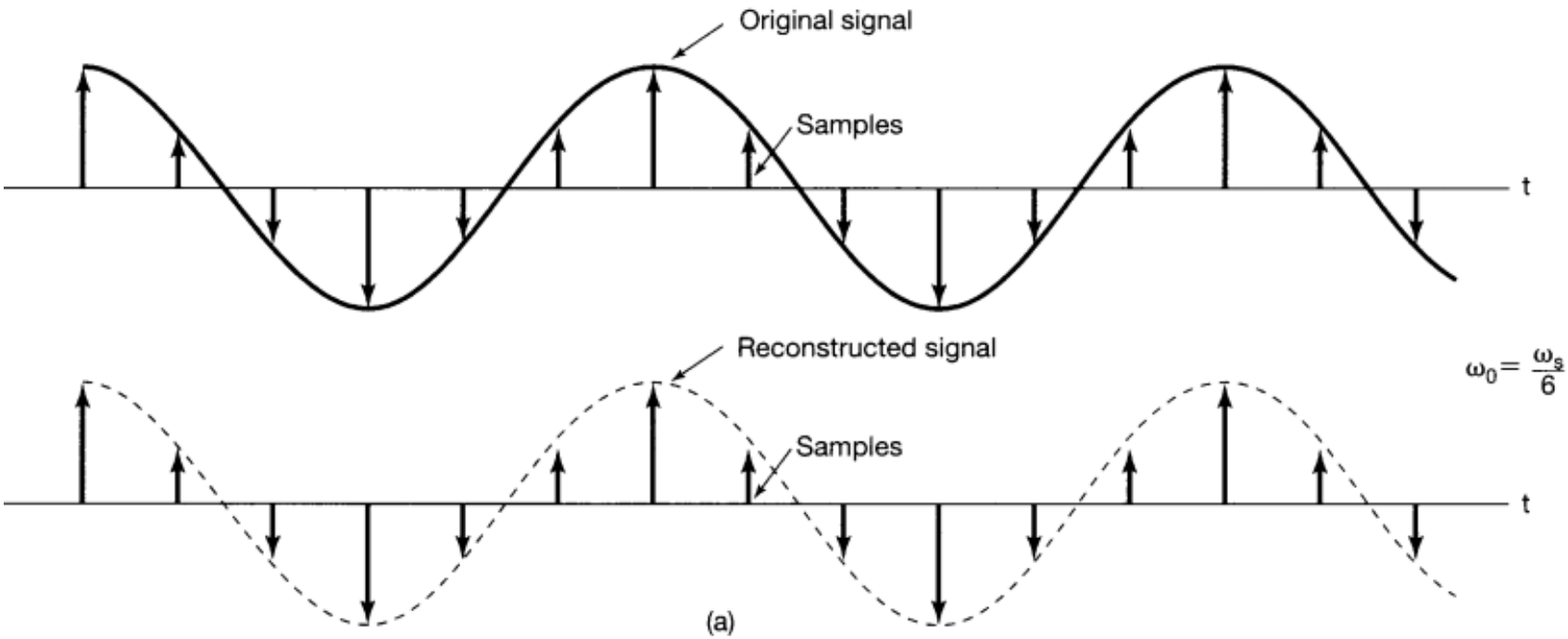


Image credit: Oppenheim 7.3

Aliasing

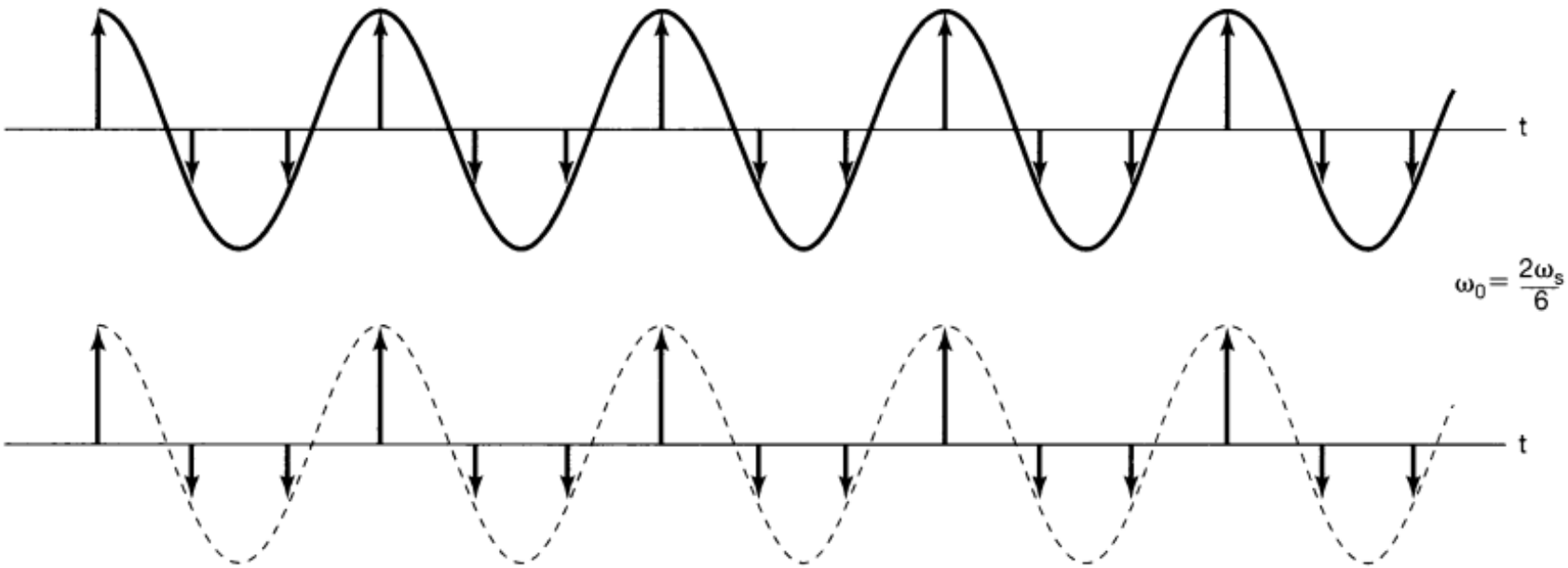


Image credit: Oppenheim 7.3

Aliasing

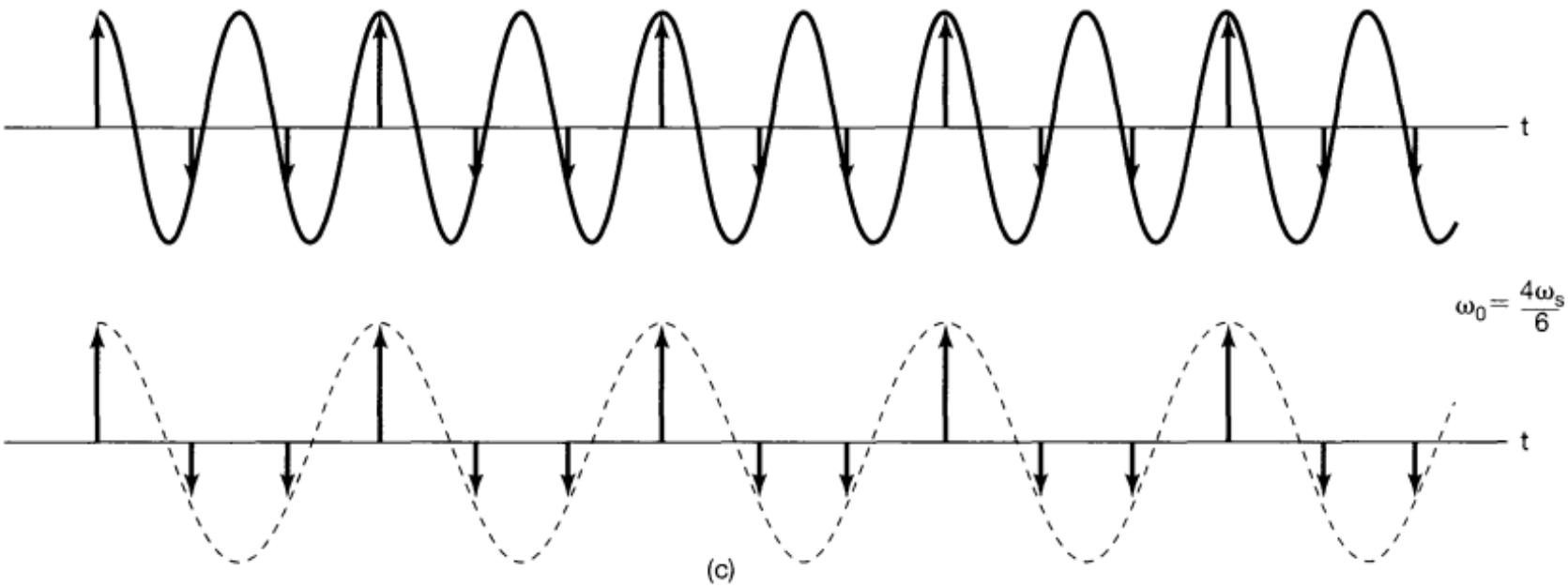


Image credit: Oppenheim 7.3

Aliasing

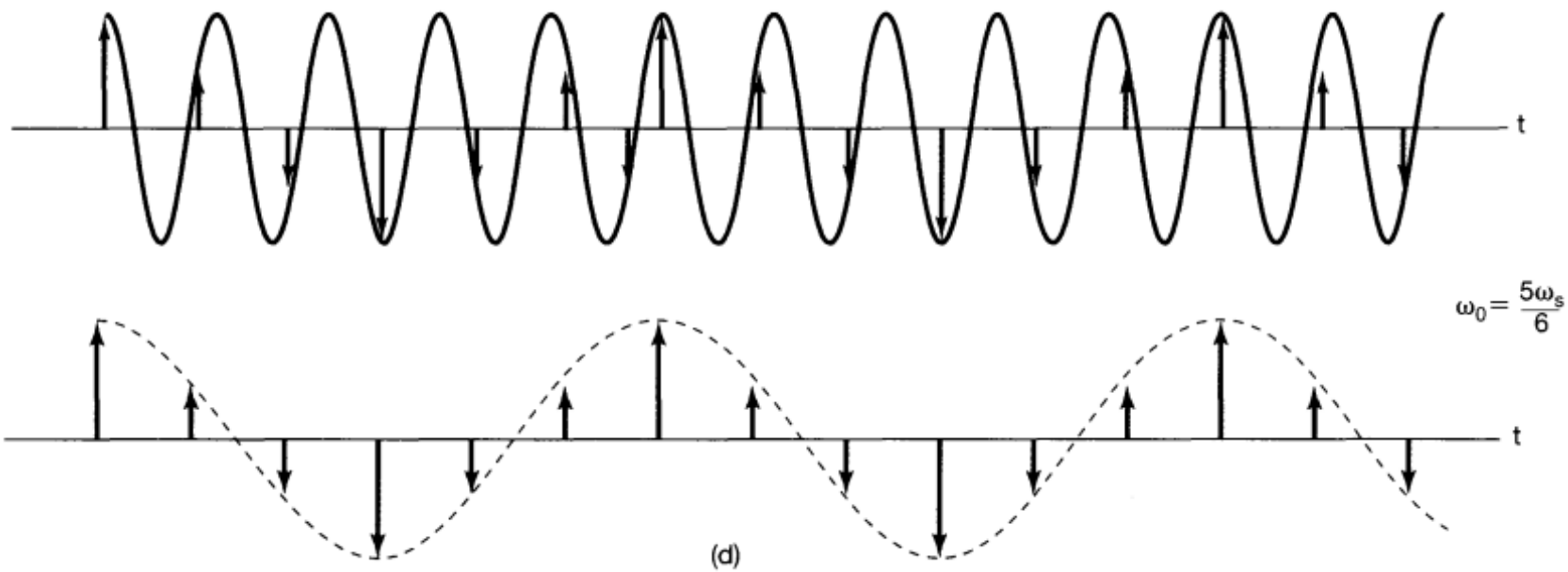


Image credit: Oppenheim 7.3

Simulation

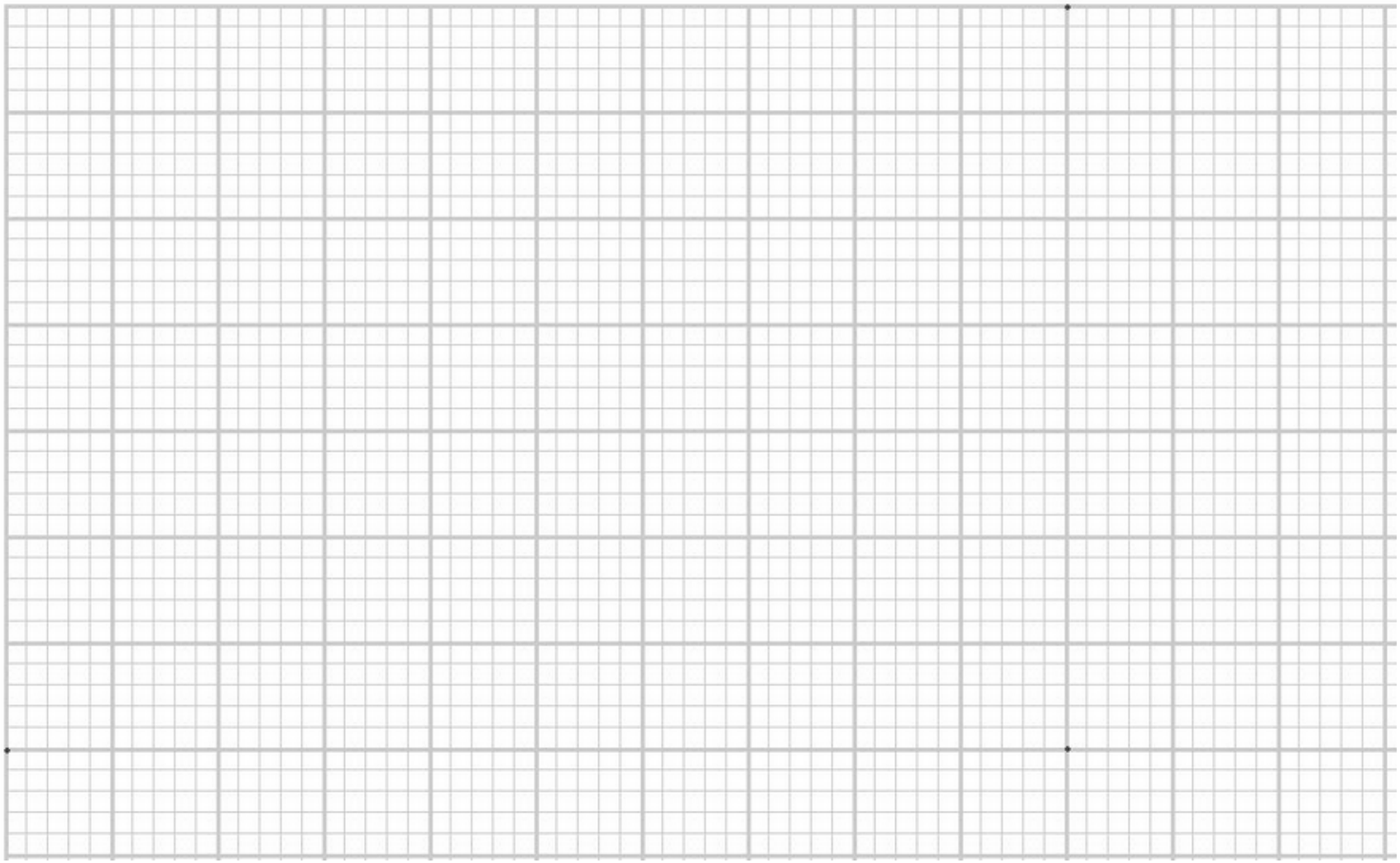
`https://visualize-it.github.io/stroboscopic_effect/simulation.html`

Two aspects to consider here:

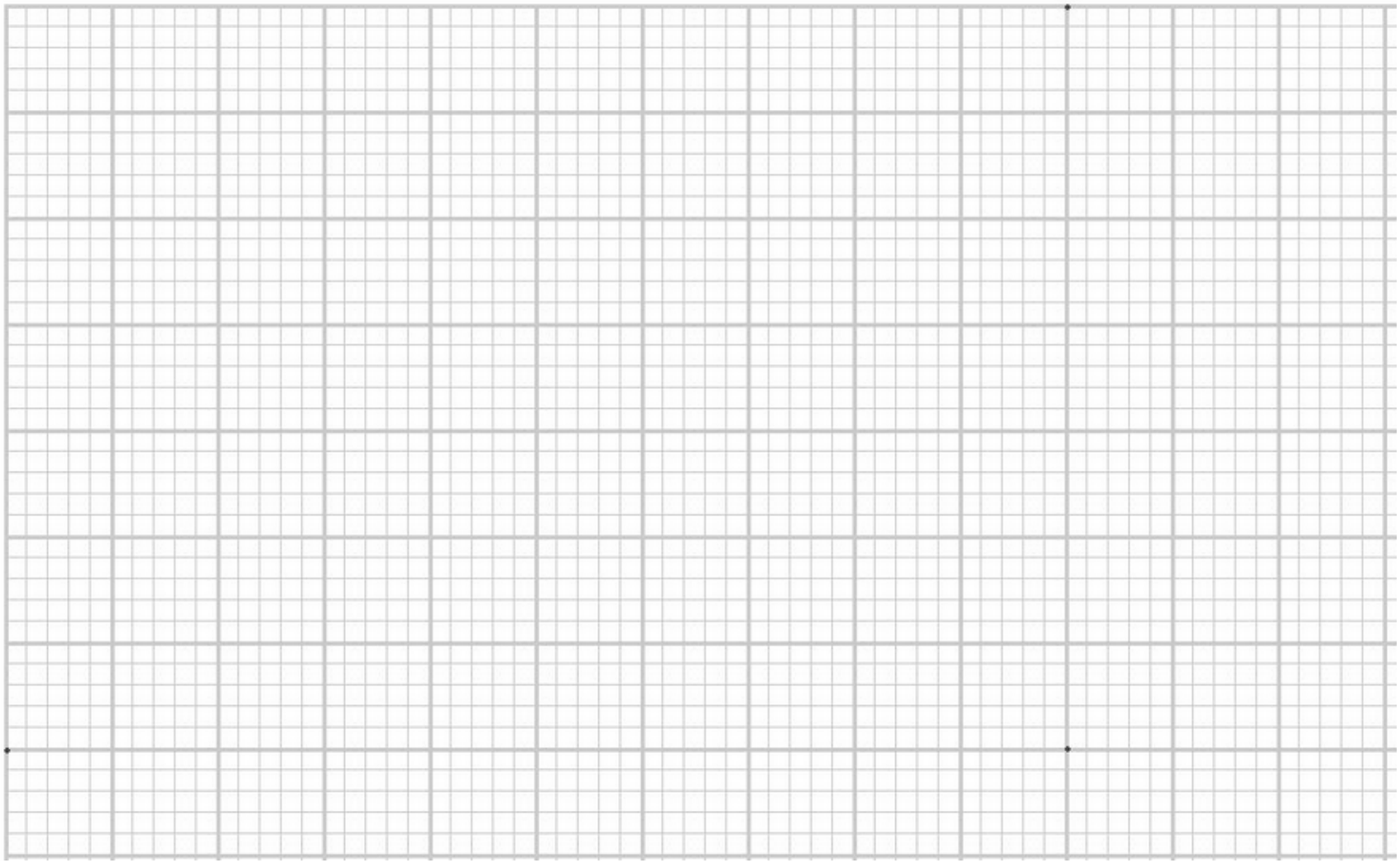
- Why does the interpreted frequency *decrease* as the true frequency increases?
- Why does it it look like it goes *backwards*?

We can understand both by looking at the spectra.

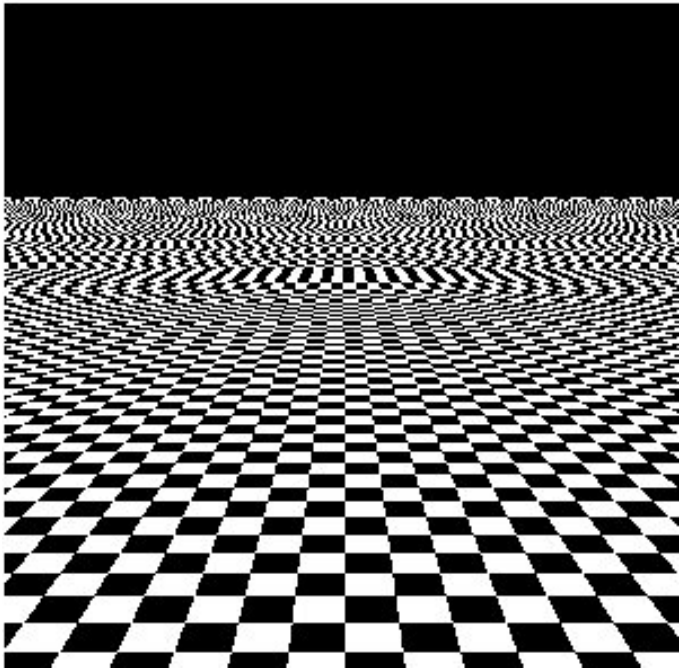
Frequency misattribution



Backwards-ness



Real-world examples



Fun on your own: read up about Moiré patterns, and various **anti-aliasing** techniques that are used in music/images/games!

Image credit: `https:`

`//textureingraphics.wordpress.com/what-is-texture-mapping/anti-aliasing-problem-and-mipmapping/`

For next time

Content:

- DT processing of CT signals
- Sampling in discrete time
- Decimation/interpolation

Action items:

1. Find examples of sampling and aliasing in real life.
2. Watch TV / movies or play games at different frame rates to study the difference.
3. Enjoy the reading break!

Recommended reading:

- From this class: Oppenheim 7.0-7.3
- Suggested problems: 7.1-7.6, 7.21, 7.25
- For next class: Oppenheim 7.4-7.6